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RESEARCH

Increasing isotropy and predictability through electroplating of tensile properties of polyamide 12 parts manufactured by selective laser sintering

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Abstract

Additive manufacturing is rapidly evolving and finding a place in nearly every industry. Recent advances make selective laser sintering especially well positioned to rapidly create complex parts with material properties capable of withstanding rigorous end use applications. However, these properties still pale in comparison to their metal powder bed fusion counterparts and can be plagued by anisotropy due to print orientation. This can be especially true for large parts with relatively thin structural walls. Combining selective laser sintering with metal electroplating has shown promise to increase both strength and isotropy. While traditional metal–polymer hybrid composites have struggled to maintain consistent properties across varying geometries, this work demonstrates control over material properties at scales previously not possible, enabling a new class of lightweight structures with improved survivability. This novel research has experimentally found the tensile properties and isotropy of different electroplating coatings and investigates techniques that can be used to increase coating adhesion and uniformity over the rough non-conductive printed part surface. It is shown that though both the printed and electroplated material are individually anisotropic, the combination of printing and subsequent electroplating can improve overall build direction isotropy. This research represents a shift in additive manufacturing by transforming what was once considered a limitation of SLS printing—thin-wall anisotropy—into an opportunity for creating customized material properties through strategic metal deposition but also opens up new design spaces where minimum weight and minimum volume design trajectories are needed.

Highlights

- Orthotropic tensile anisotropy characterization electroplated PA12 sintered parts.
- Incorporating electroplating into design for AM thin wall parts to tune stiffness.
- Characterizing the difference between electroplated thin and thick PA-12 samples.
- Analysis of electroplating coating impacts on growth and transverse directions.
- Hybrid disparate phase material characterization.

Keywords Additive manufacturing (AM) · Selective laser sintering (SLS) · Polyamide-12 (PA12) · Orthotropic · Electroplating · Hybrid materials · Cellular solid

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Abbreviations

AM	Additive manufacturing
ANOVA	Single factor analysis of variance
cc	Cubic centimeter
Cu(S)	Structural copper
D_oA	Degree of anisotropy
DOE	Design of experiments
DFAM	Design for additive manufacturing
E_A	Axial elastic modulus
EDM	Electron discharge machining
EBSD	Electron backscatter diffraction
g	Gram
GD	Growth direction
ISDG	Interferometric strain/displacement gauge
MJF	Multi jet fusion
mm	Millimeter
NDPA	Nickel deposit with a platinum anode
PA12	Polyamide-12
ROM	Rule of mixtures
SLA	Stereolithography
SLS	Selective laser sintering
TD	Transverse direction
TNS	Traditional nickel sulfamate
UTS	Ultimate tensile strength
ε_f	Strain at failure
σ_y	0.2% Yield strength
μ-MTS	Micro-material testing system

1 Introduction

Polymers have found application in nearly every industry [1] and additive manufacturing (AM) has been a critical manufacturing method used to expand polymer applications. Powder bed fusion (PBF) is an AM method with proven success in creating complex parts and within PBF selective laser sintering (SLS) is an established method capable of creating end use polymer parts [2, 3]. Within this manufacturing method, polymamide-12 (PA12) is one of the most cost-effective materials due to its reliability, durability, and chemical resistance [4, 5]. PA12 is specifically well suited for laser sintering because of the relatively low sintering power requirements, high material refresh ratios, and ability to print without an inert chamber [6, 7]. SLS can also produce highly intricate, nested, and lightweight parts without the use of support material at extremely competitive cost compared to traditional manufacturing methods.

Even though SLS with PA12 can produce end use parts at a fraction of the price and weight of metal PBF, the strength of PA12 parts cannot compete with metal PBF. Metal parts often have higher wear resistance, thermal resilience, and electrical conductivity. Polymer parts are also susceptible to UV, hygroscopicity, and degradation when exposed to other liquid media [8, 9]. Another common detrimental biproduct of PA12 PBF is anisotropy of the part based on build orientation and part thickness [10–14]. Electroplating is a well-established procedure that is poised to not only mitigate many of the undesirable environmental effects on polymers but also increase the strength of polymers [15–19]. Electroplating continues to play a more critical role in AM with research and implementation skyrocketing in recent years [20, 21]. Much of the progress in combining AM and electroplating (often referred to as hybrid AM or HAM) involves selectively depositing conductive material to make 2D and 3D circuit boards [22–25]. Electroplating also combines the benefits of low weight polymers with the benefits of metal surfaces due to the minimal thickness of the metal coating [15, 16, 26, 27]. With such broad advantages of electroplating, this technology has found use in nearly every industry to include automotive, electronics, aviation, oil and gas, air conditioning, toy and jewelry manufacturing, and national defense [21]. None of the found research employs electroplating with AM to address issues of AM print anisotropy. Simonson et al. did investigate using electroplating to decrease layer-to-layer debonding of fused filament parts but found electroplating polylactic acid parts manufactured through fused deposition modeling actually weakened the tensile properties by approximately 20% [28]. Zarybnicka et al. demonstrated similar results with sintered PA12 if additives are added to the powder prior to sintering which can result in a decrease of tensile properties of the final part and little effect on part anisotropy due to print orientation [29]. This research investigates novel techniques to incorporate electroplating into the PA12 PBF process to address polymer environmental degradation, strength, and anisotropy issues while maintaining total manufacture cost and end part weight at a fraction of the metal counterpart. This is particularly true with thin-walled (< 1 mm) SLS structures. The present research also demonstrates the ability to use the rule of mixtures (ROM) to predict and customize material properties based on coating type and volume fraction.

Electroplating non-metallic parts holds its own challenges; however, when the parts also have rough non-uniform surfaces found on SLS parts, the challenges are compounded and often result in poor bonding at the polymer–metal interface [17, 29–31]. Baligidad et al. found poor bonding at the metal interface was detrimental to mechanical properties of the end part [8]. Methods to address the first challenge of creating a conductive material to include surface embedding conductive material in the polymer, precoating the part with an adhesive spray, and treating the surface with an electroless coating have all been met with mixed success [8, 16, 17, 19, 22, 24]. Creating a relatively smooth surface is the second common challenge and attempts to mitigate this are often done through mechanical methods chemical changes or a combination of these two methods

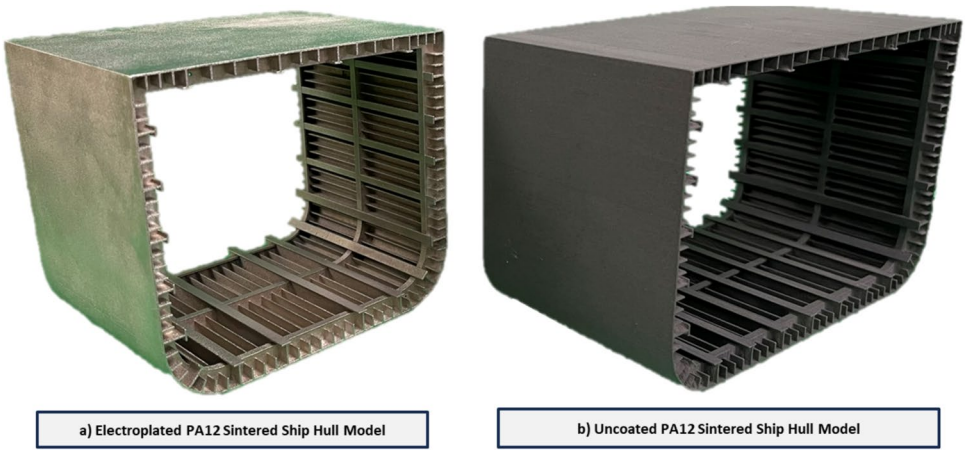


Fig. 1 Hydro-elastic ship model fabricated using SLS with 0.8 mm PA12 wall thickness and coated with a three-part electroplating process

[17, 30–32]. Some studies have also found that the adhesion of a metal coating on laser sintered PA12 and the continuity and uniformity of follow on metal coatings was increased by starting with a thin layer of Cu [8, 17]

This research has employed a unique order of electroplating to achieve uniform adhesion and thickness at the material interface. The research community and some commercial entities [16, 30] have done extensive testing of electroplating AM parts and have found that SLS prints generally have a great deal of surface variability. This can be seen to a lesser degree with other AM printing methods, such as stereolithography (SLA) and multijet fusion (MJF). A very simple screening test is to weigh parts prior to plating to look for porosity. The weight in grams of a part divided by the part's volume, which can be read from the 3D model, gives an indication up front if the part is porous. A dimensional check on the printed part must also be done to ensure the printed part size matches the 3D model. The part weight in grams (g) divided by its volume in cubic centimeters (cc) should be very close to 1 for PA12 parts. PA12 parts whose density is below 0.95 g/cc often will have issues of staining and will likely be weaker as well. It is also essential that there are no residual powders on the parts.

While traditional metal–polymer hybrid composites have struggled to maintain consistent properties across varying geometries, this work in electroplating SLS parts demonstrates control over material properties at scales previously not possible, enabling a new class of lightweight structures with improved survivability. The ability to tune material properties through detailed electroplating processes bridges the long-standing gap and fills in gaps in the materials selection in mechanical design space between polymer and metal AM parts. This novel research represents a shift in AM by transforming what was once considered a limitation of SLS printing—thin-wall anisotropy—into an opportunity for creating customized material properties through strategic metal deposition, but also opens up new design spaces where minimum weight and minimum volume design trajectories are needed.

The results of this study can be applied to many different industries to improve, and tune material properties of printed parts to fit a specific application. One of those applications is structurally scaled ship models and two papers by Ibrahim et al. investigated applying the discussed techniques to create structurally scaled ship models for hydro-elastic model test experiments [33, 34]. This type of experiment is geared toward measuring global design loads on ships and their structural response [35, 36]. To achieve the desired scaled structural stiffness, the models must have relatively large overall dimensions with very thin thickness (0.8 mm) for the structural members (Fig. 1). Electroplating the model using the methods discussed in this paper results in increasing the model structural stiffness while maintaining the very thin thickness of structural members. Furthermore, electroplating the model creates a watertight layer to eliminate moisture absorption during testing in a towing tank. This paper includes calculations to predict performance of such models enabling both the geometry and the stiffness to be appropriately scaled. This case study has proven the feasibility to combine SLS PA12 thin-wall structures with electroplating to decrease anisotropy while increasing stiffness and environmental resilience.

2 Materials and methods

2.1 Design of experiments

A design of experiments (DOE) was developed to analyze electroplating effects on the tensile properties of PA12 parts manufactured using SLS. The DOE focused on establishing methods to compare the isotropy of plated (electroplated) and unplated SLS parts. Tensile characterization from a previous study by Slager et al. was leveraged to set the baseline for unplated tensile data [10]. To achieve a direct comparison

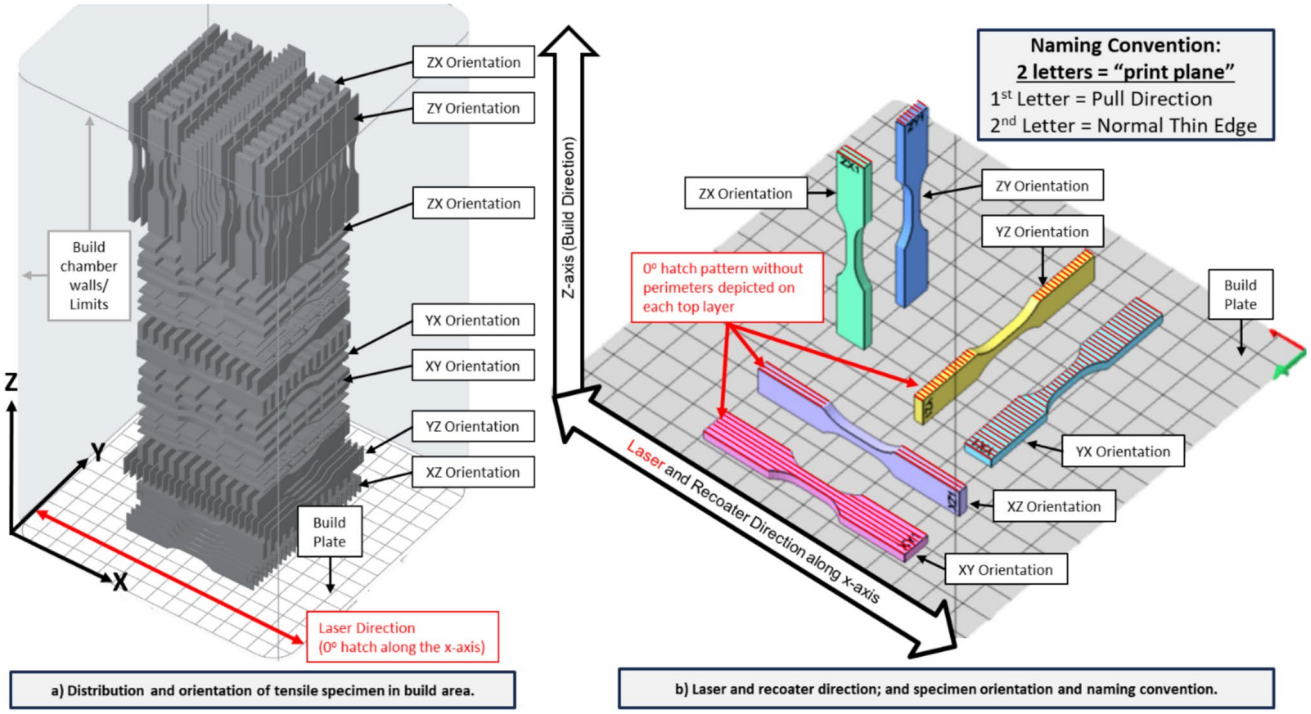


Fig. 2 SLS orthotropic tensile characterization design of experiments; **a** distribution and orientation of all D638 Type-V specimens in a single print; **b** expanded view highlighting laser and recoater direction and specimen print orientation and naming convention (Design of Experiments, naming convention, and figure adapted from Slager et al. [10])

to the unplated study, tensile specimens printed during the initial study were plated and analyzed. The printed batch included 144 ASTM-D638 tensile specimens [37] with 24 specimens in each orthotropic orientation (Fig. 2a). Each orientation included 12 standard specimens (3.2 mm thickness) and 12 thin specimens (0.8 mm thickness). The standard and thin samples were of the same dimensions with the exception of specimen thickness. To further isolate changes in material properties to orientation, thickness, and plating, all specimens were printed in the inner 60% of the build chamber and a constant x direction hatch pattern without perimeters was used (Fig. 2b). A naming convention was adopted to distinguish between print orientation, thickness, and plating. A two-letter identifier described the orientation, where the first letter identified the axial (pull direction) and the second letter identified the transverse direction of the tensile specimen. The specimens were further classified as 'standard' or 'thin' to denote thickness, and 'unplated' or 'plated' to indicate electroplating status.

2.2 Additive manufacturing

All specimens were printed in a single build using a Fuse 1 SLS printer manufactured by Formlabs (Formlabs, Somerville, MA). The Fuse 1 utilizes a 10W fiber laser with a 200 microns spot size and an ambient air build chamber with temperatures up to 200° C. Specimens were orientated and arranged within the build area using Autodesk (Autodesk, San Francisco, CA) Netfabb AM software (version 2024). The build file was sliced using Formlabs PreForm slicing software (version 3.29). Laser power and hatch pattern can affect the mechanical properties of SLS parts [38]. PreForm minimizes mechanical property changes by adjusting laser speed based on cross-section through a patent-pending "Surface Armor" technology. Print settings matched a previous study by Slager et al. [10] to achieve a direct comparison to uncoated material properties. A layer height of 0.11 mm and a constant x-direction hatch pattern without perimeter matched the study done by Slager et al. (Fig. 2b). Laser power, recoater direction, and temperature profile were also held constant for all samples to further isolate the effects of print orientation and match the previous uncoated study by Slager et al. (Fig. 2) [10].

2.3 Electroplating

The electroplating process generally followed ASTM B604-91 (Standard Specification for Decorative Electroplated Coatings of Copper Plus Nickel Plus Chromium on Plastics) [39]. Electroplating requires the cathode (the part being plated) to be conductive. This was accomplished by first preparing the surface to be electroless plated with a thin

Table 1 Bath chemistry for electro deposition

Material deposited	Copper	Nickel	Nickel
Bath type	Copper sulfate	Nickel sulfamate	Nickel sulfamate
Anode	Copper	Sulfur depolarized nickel	Sulfur depolarized nickel
Secondary anode	None	None	Platinum plated titanium
Additive	None	Sulfamic acid (pH adjustment)	Nickel carbonate (pH adjustment)
Surfactant	Polyethyl glycol	None	None
Anti-pitter	None	Sodium laurel sulfate	Sodium laurel sulfate
pH buffer	None	Boric acid	Boric acid

layer of autocatalytic Ni which was found to provide better bonding of the subsequent electroplating layers than covering the surface with a conductive paint. Surface preparation and conductive coating methods generally followed ASTM B727 (Standard Practice for Preparation of Plastics Materials for Electroplating) [40]. Excessive powder removal from the surface of the specimen was critical to ensure uniform application of the conductive seed material. Abrasive spray was used to ensure surface consistency and uniform application of seed material. Small copper wires were inserted into the grip section of each specimen to facilitate application of the conducive seed material and conductive mounting onto the electroplating racks. Residual soils were removed via a soapy bath followed by activator-catalyst soak. An additional accelerator tank was used to remove access activator residue. Final conductive seed material is a 1-micron coating of electroless nickel applied in an electroless nickel bath.

Electroplating baths followed ASTM B832-93 (Standard Guide for Electroforming with Nickel and Copper) [41]. This procedure was modified in accordance with the Tool and Engineering Handbook (CH9: Coatings, Metal Deposition) [42] to adapt the process for structural parts and applications. Growth conditions in the electrodeposition process can directly impact the resultant material's physical and mechanical properties [43–50]. For this work, the current density, voltage, temperature, and flow rate were all precisely controlled and not varied. Specifically, the current density was held constant at 6.46 mA/cm^2, while the voltage was a free variable allowed to change from 0.4 to 2 volts. The voltage is very dependent on the load in the tank (size and number of samples in the electroplating bath). The bath temperature was 43 degrees Celsius with a pH of 3.9-4.2 for nickel sulfamate and no less than 24 degrees Celsius and pH well below 1 for the copper sulfate bath. A high throw copper bath with a high concentration of sulfuric acid (≥10%) and a relatively low concentration of copper sulfate was used to increase conductivity to plate more evenly with complex geometries and help with rougher surfaces.

This resulted in a deposition rate of 6 microns per hour for the electrodeposited materials. In total, three electrodeposition baths were investigated in this study: one for growth of copper and two for growth of nickel. The general bath chemistry can be found in Table 1.

ASTM B689 (Standard Specification for Electroplated Engineering Nickel Coatings) breaks down nickel coatings into Type 1, Type 2, and Type 3 coatings. The major difference between the two nickel baths used is that the nickel deposited with a platinum anode (NDPA) bath, which falls within the Type 2 definition of the ASTM B689 standard, has a supplemental anode which is a titanium wire coated in a fine layer of platinum. This coated titanium wire is required for parts with deep cavities to enhance the electric field of the solution in recessed areas. Platinum is used to resist oxidation and is generally considered non-consumable in the electroplating process. Over-plating at sharp points can be caused by a spike in the electric field [50]. However, part placement, spacing, side shields, and thieves are used to reduce these high fields and achieve uniform plating. Additionally, a reduction in the electric field can occur in high-aspect-ratio holes due to geometric constraints reducing the throwing power of the electric field [50]. To combat this phenomenon, a platinum wire mesh is used on a separate power supply to make the bath's electric field more uniform around complex parts. A Traditional Nickel Sulfamate (TNS) bath, which falls within the Type 1 definition of ASTM B689 standard, was also used. The Type 1 (TNS) bath, which is never run with insoluble supplemental anodes, tends to drift up in pH, so pH is adjusted down using sulfamic acid additions as needed. The Type 2 baths, when run repeatedly with insoluble anodes, tend to drift down in pH, so pH is adjusted using nickel carbonate.

2.4 Materials

2.4.1 Additive manufacturing materials

Formlabs branded PA12 powder [51] was used for all SLS manufacturing. This powder is specifically developed for the Fuse Series printers and most powder properties remain proprietary. To decrease waste during the SLS manufacturing powder unsintered used powder from previous SLS prints (recycled powder) is typically mixed with fresh

unused powder. Refresh ratio (percentage of fresh powder to total powder) must be taken into account when evaluating mechanical properties, because powder coalescence, part porosity, and crystal structure may vary based on the amount of recycled powder [52]. Refresh ratios of 15–50% are often used to manufacture end use parts and characterize mechanical properties [29, 53, 54]. A 30% refresh is recommended by Formlabs [7, 55] and was used in this study.

2.4.2 Electroplating materials

Electroplating materials selected for coating the sintered PA12 tensile specimens were sequentially layered equal parts of structural copper [Cu(s)], a ductile nickel bath (Traditional Nickle Sulfamate [TNS]), and hard nickel bath (Nickle deposit with a platinum anode [NDPA]). Nominal plating thickness was 0.0045" (114 microns) based on total current (amp-hours/foot^2) and time in each bath. A separate bath was used for each material. Copper was deposited first as electroplating tests on SLS parts demonstrated its superior tolerance to the relatively rough powder surface. The thickness of each material layer was measured using two methods. First, by measuring the cross-section of the gauge section before and after electroplating. Second, by weighing each specimen before and after plating and calculating the plated material volume using the known density of each material and assuming uniform coating thickness. The thickness of each material was calculated by dividing total plate thickness by number of material baths (assumed equal thickness of each material).

2.5 Mechanical testing of plating material

Material properties of these coatings were experimentally determined using microtensile specimens. Traditional samples for characterizing materials are insufficient for determining the mechanical response of electrodeposits, since film properties are coupled to thickness and evolution of microstructure. If the standard ASTM D1708-02A sample, which is 38 mm x 15 mm with a gage section of 5 mm by 3 mm, was used, property evolution due to the manufacturing process would be lost, because the thickness of the sample is beyond the evolution of the grain texture [56]. Zupan et al. developed a novel microsample that enables investigation on a much smaller length scale [57, 58]. This study used a similar testing geometry with an overall sample dimension of 3×1 mm and a gage section of 250 x 250 μm [59–61]. Characterization of the evolution of the coating is possible, because this sample allows independent measurements of the properties in both the growth direction and the transverse direction. Figure 3 shows the micro-material testing system (μ-MTS) (Fig. 3a) and a micro-tension sample (Fig. 3b). A non-contact laser-based interferometric strain/displacement gage (ISDG) was used. Strain rates of 1×10^{-4} per second were used for all samples.

2.5.1 Microsample extraction/ fabrication

Microsamples were extracted from nickel and copper electrodeposited films approximately 4 mm thick (Fig. 4). The samples were cut using plunge electron discharge machining (EDM). This process has been shown to yield samples with a repeatable accurate cross-sectional tolerance to within 0.1 μm. The samples were extracted in two specific orientations: the transverse direction (TD) and the growth direction (GD). This allows characterization of how the material properties change with respect to orientation.

2.6 Mechanical testing under tensile loading

Tensile testing was conducted using ASTM D638-22 Type-V [37] uniaxial tensile test standards for all samples with a crosshead rate of 1 mm/min. Prior to testing, all samples were maintained in a 73.4 ± 3.6°F; 50 ± 10% relative humidity laboratory environment for at least 40 h. The samples were then measured at three distinct cross-sectional planes for width and thickness before testing in accordance with ASTM-D618 [62].

The tensile testing was performed using an Instron (Instron, Norwood, MA) Model 4466 screw driven test frame with a Model 2518–806 load cell (verified tensile range of 1.00–200lbf) to apply and resolve force. Specimen grips were reinforced with a single layer of 120-grit Norton Metalite (Saint-Gobain Industries, Staverton, Gloucestershire) abrasive paper. Due to the use of digital image correlation for longitudinal deformation measurements, the specimen gauge section was painted. Following specimen preparation, the specimens were mounted in the Instron by centering and restraining each specimen within serrated parallel face grips with sufficient applied pressure to preclude slippage during force application. A non-contacting digital imaging correlation system was used to obtain longitudinal deformation measurements. The system consisted of an Imetrum (Imetrum Limited, Bristol, UK) Model SC-2 processor and an Allied Vision (Allied Vision, Thuringia, Germany) Mako Model G-192B PoE camera with Imetrum Video Gauge Version 5.4.8 (Build 6128) software.

2.7 Data collection and calculations

Load, axial displacement, and transverse displacement were collected for each type of electroplating material [Cu(s), TNS, and NDPA] in both GD and TD; and for three sintered and coated specimens of each orientation and thickness. The following values were analyzed: Elastic Modulus (E_A), 0.2%

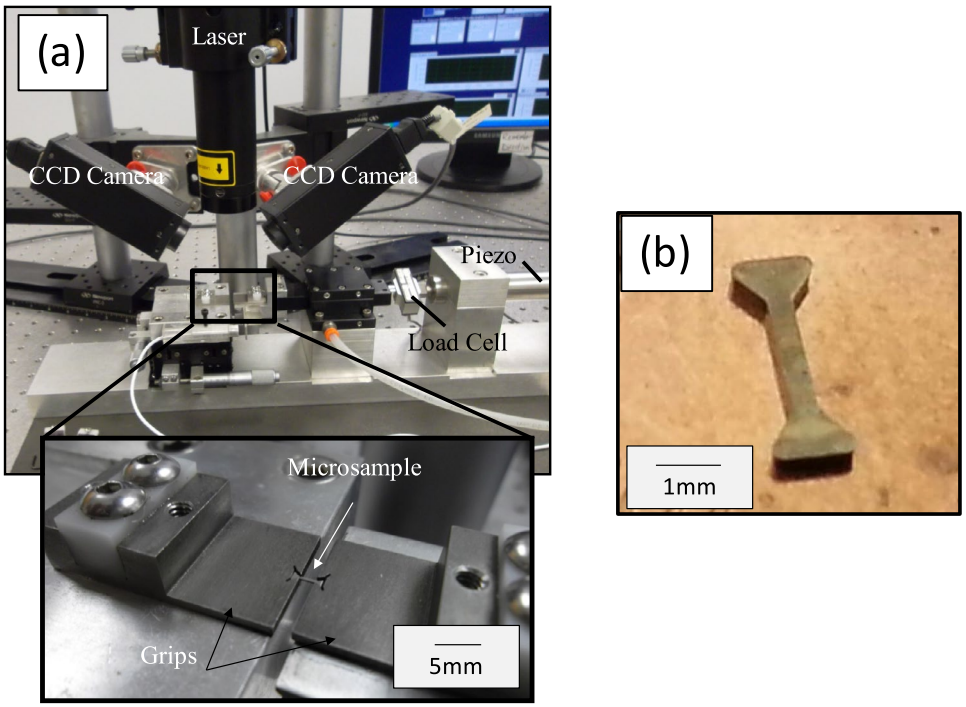


Fig. 3 Overview of **a** µ-MTS testing system and **b** micro-tension sample used to characterize the electrodeposited metals

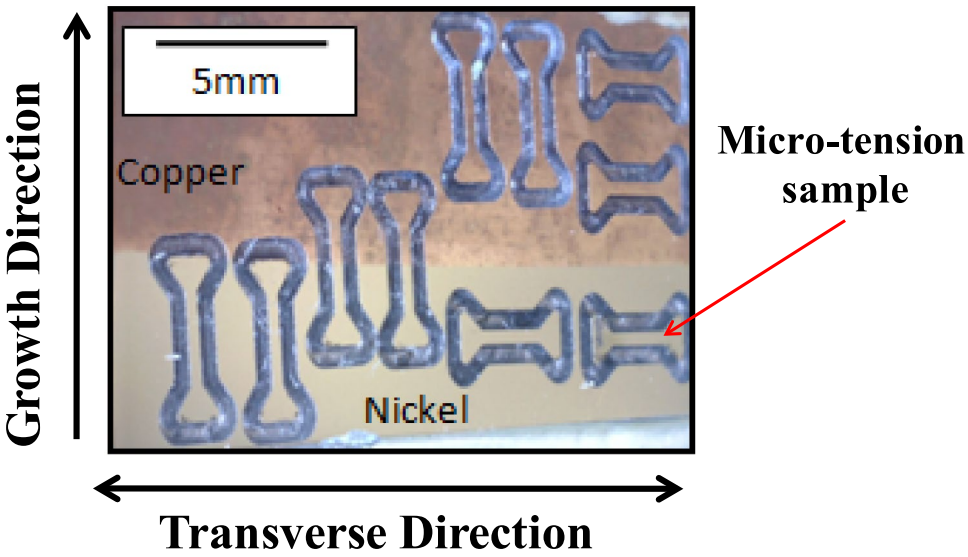


Fig. 4 Grid of microsamples electron discharge machined from a bulk deposit. Each layer is approximately 4 mm of growth

Yield Strength (σ_y), Ultimate Tensile Strength (UTS), Strain at Failure (ε_f), and degree of anisotropy (DoA). Single factor analysis of variance (ANOVA) with a significance level of 0.05 ($\alpha = 0.05$) calculations were done on E_A, σ_y, UTS, and ε_f, to determine possible statistical difference between each print orientation within each specimen type (uncoated and coated standard and thin). If the ANOVA calculations indicated a statistical difference between orientations, an F test for variance was completed and P values computed to determine appropriate variance assumptions (equal or unequal). Each possible orientation pair was then analyzed with a two-tailed t test to determine possible statistical difference of E_A, σ_y, UTS, and ε_f between print orientations (assuming changes in material properties could either increase or decrease and equal variance assumed for P values > 0.05 while unequal variance assume for P values ≤ 0.05).

ROM equations were used to determine the predicted strength of each plated specimen [16]

$$E_{plated} = E_{SLS_PA12}V_{SLS} + E_{Cu(s)}V_{Cu(s)} + E_{TNS}V_{TNS} + E_{NDPA}V_{NDPA} \quad (1)$$

$$\sigma_{plated} = \sigma_{SLS_PA12} V_{SLS} + \sigma_{Cu(s)} V_{Cu(s)} + \sigma_{TNS} V_{TNS} + \sigma_{NDPA} V_{NDPA}, \quad (2)$$

where E_{plated} is the predicted elastic modulus of the plated tensile specimen, E_{SLS_PA12} is the experimentally determined elastic modulus of the unplated printed tensile specimen, V is the volume fraction of each material, and $E_{Cu(S)}$, E_{TNS}, and E_{NDPA} are the elastic moduli for each plating material. In Eq. (2), σ_{plated} is the predicted strength (either ultimate or yield) of the plated tensile specimen, σ_{SLS_PA12} is the experimentally determined strength of the unplated printed tensile specimen, and $\sigma_{Cu(S)}$, σ_{TNS}, and σ_{NDPA} are the strength for each plating material.

The D_oA was also analyzed for all material properties evaluated during mechanical testing (E_A, σ_y, UTS, and ε_f). The influence build orientation had on material properties was calculated using Eq. (3)

$$D_O A_{orientation12} = \frac{\overline{M_orientation_1} - \overline{M_orientation_2}}{\overline{M_orientation_{avg}}} x100\%. \quad (3)$$

The calculations matched previous work done by Slager et al. [10] on uncoated specimens to facilitate a direct comparison with the coated specimen results where $D_oA_{orientation12}$ represented the D_oA at two different orientations (orientation "1" and "2"). $M_orientation_1$ represents the first chosen orientation (XY, XZ, YX, YZ, ZX, or ZY) and $M_orientation_2$ represents the second orientation chosen for comparison (XY, XZ, YX, YZ ZX, or ZY). $M_orientation_{avg}$ is the average of the two material properties at the two different orientations. Alcisto and Sindinger [11, 63] also used a similar method to describe the degree of anisotropy but did not normalize by the average value in the denominator. The anisotropic effects of orientation at a specific thickness are calculated using the numerator values in Eq. (3).

2.8 Fracture characterization

To evaluate the impact of sample thickness, coating of samples, and print orientation, microscopy of the fractured tensile surfaces was performed. A TESCAN (Brno, Czech Republic) MIRA 3 Scanning Electron Microscope (SEM) was used to obtain images. All of the fracture surfaces of the PA12 coated samples were sputter coated with a thin (~5nm) coating of gold to create a conductive pathway and support effective SEM imaging. Images were obtained with the SEM operating in secondary electron mode at 5 keV, and images were used to examine particle coalescence, visualize coating layer, and analyze surface morphology.

Additional microscopy of two non-fractured coated samples was performed to evaluate for any differences in the coating to PA12 interaction. The samples were sputter coated and imaged in the same manner as all of the fractured samples.

3 Results and data analysis

3.1 Electroplating coated material results

A representative stress vs strain curve for each of the electroplating processing cases is shown in Fig. 5. The axes scales for all plots are consistent to allow for direct comparisons of mechanical properties. From these experimental results, mechanical properties E_A, σ_y, UTS, and ε_f were extracted for the GD and the TD. The solid lines in the curves represent data measured from laser ISDG. The final strain-to-failure measurement was determined using optical microscopy after completion of the experiment by direct measurement of the as-deformed sample cross-section. A dotted line connects the last measured ISDG point to the final as-measured strain-to-failure point. The dotted line is only an appraisal of the plastic work hardening deformation response to the strain-to-failure point. The dotted line is provided to demonstrate that additional strain has accumulated before rupture. The determined mechanical properties and population standard deviations are presented in Table 2.

3.2 Electroplating coated material microscopy results

The grain structure for the TNS deposited nickel was determined to be 786 nm, while the NDPA nickel has a grain size of 74 nm. This grain reduction is consistent with strengthening due to a Hall–Petch type mechanism in which the strength scales proportional to the inverse square root of the grain size [3, 31]. Strength is increased, because the path of motion of the dislocations is reduced, such that once they glide across the smaller grain center, they are impeded or pinned by the grain boundaries.

Figure 6 shows the inverse pole figures and grain size distributions for the NDPA nickel starting from the substrate through 1.5 mm of growth. A series of 3 scans were taken to characterize the NDPA nickel at various lengths from the substrate. The first scan was taken in the first 5–15 µm of the sample, the second scan was taken at 1 mm, and the third scan was taken at 1.5 mm above the substrate. Each scan was enhanced using CI standardization followed by one iteration of grain dilation. The grain size and morphology of the NDPA nickel material evolves in size as you traverse the growth direction. In the first 5–15 µm, the grain size is measured as small as 20 nm for the average grain size in the scan. The grain morphology evolved from equiaxed to slightly elongated grains, with the major axis aligned with the growth direction. This increase in grain size along the

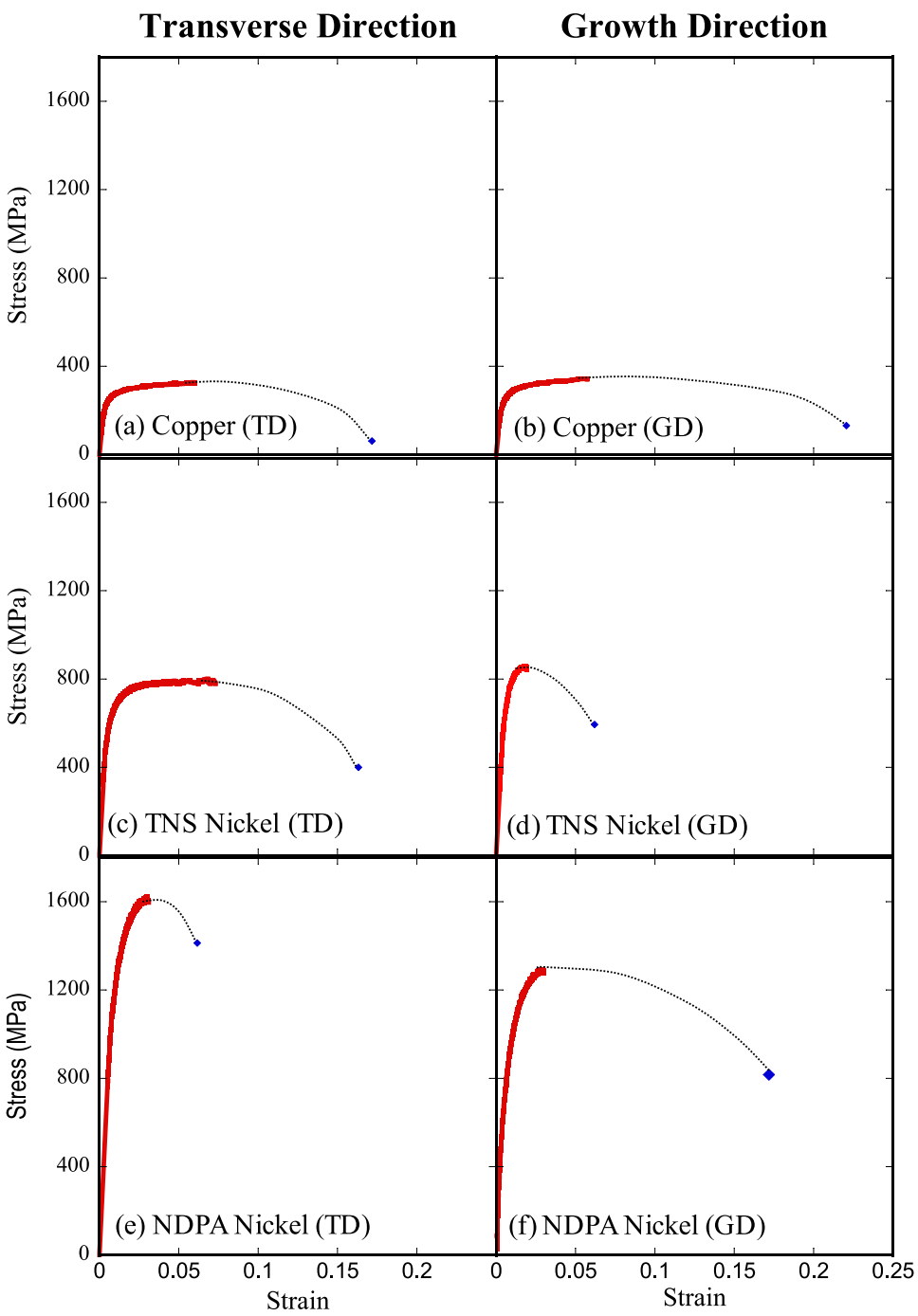


Fig. 5 Representative curve for each of the microsamples tested

growth direction correlates with decreased strength in the metallic coating as thickness increases. The elongated grain structure introduces material anisotropy. The grain elongation direction oscillates between 10 and 15 degrees relative to the growth direction. This alternating grain orientation promotes grain rotation as a primary deformation mechanism, resulting in reduced strain-to-failure as energy is dissipated through rotation rather than grain boundary sliding. The inverse pole figures also give light into the directionality of the electrodeposits. The growth direction for NDPA nickel trends toward < 201 > texture directions. The texture relaxes slightly toward the < 100 > direction as the coating grows. This texture inside of the material can change the plastic response and result in a higher yield point for the growth direction.

Texture was also analyzed for the TNS to see if the trends followed for both electrodeposited materials. Figure 7 shows the texture for both NDPA nickel and TNS nickel. In this figure, the same scale is used for both samples and the scale was switched to linear to make direct comparisons of the

Table 2 Properties of electroplating deposits

	Direction	E (GPa)	σ_y (MPa)	UTS (MPa)	ε_f (%)
Structural copper	Growth (GD)	62.5	230.1	332.2	22.1
	SD	5.44	16.9	15.6	2.2
	Transverse (TD)	60.3	253.4	327.7	17.2
	SD	3.25	6.1	0.71	1.8
Nickle TNS	Growth (GD)	145.8	705.1	870.3	6.1
	SD	16.1	28.2	7.8	2.3
	Transverse (TD)	165.8	607.2	861.1	16.3
	SD	39.1	7.1	1.9	1.6
Nickel NDPA	Growth (GD)	148.1	921.2	1294.5	17.2
	SD	9.7	11.1	15.1	1.2
	Transverse (TD)	146.2	1288.7	1558.7	6.2
	SD	20.1	9.8	31.8	1.1

two materials. The TNS sample appears to have minimal to no texture preference.

3.3 Coated sintered PA12 tensile test results

The mechanical testing performed on the 18 coated standard samples and 18 coated thin samples followed the same procedures previously done by Slager et al. on uncoated samples [10] to allow a direct comparison between the coated and uncoated samples. Uncoated data, that were previously published by Slager et al. [10], are incorporated in the tables (Tables 3, 4, 5, 6 and 7) and figures (Figures 9, 10, 11, 12) of these coated results to facilitate a direct comparison and discussion on the advantages and disadvantages of electroplating SLS parts. E_A, σ_y, UTS, and ϵ_f were determined and individually analyzed below. Stress–strain curves were generated for a selection of the fractured samples to provide representative graphs that allowed a comparison of the impacts of thickness and print orientation on stress and strain. A consolidated stress–strain graph showing coated samples of both standard and thin thickness is provided in Fig. 8. Follow-on graphs and tables are presented that compare the effects of sample thickness and print orientation on E_A, σ_y, UTS, and ϵ_f.

3.3.1 Axial elastic modulus (E_A)

A graphical representation of the effects of thickness, print orientation, and electroplate coating on E_A is provided in Fig. 9 and Table 3 numerically presents the same results. The uncoated data were part of a prior study [10], while the coated data represent the electroplated samples. This graph shows that the coated samples all had a significantly higher stiffness than their uncoated counterparts. In general, the coated standard samples for the XY, XZ, YX, and YZ print orientations had a slightly higher stiffness than the ZX and ZY print orientations. For the coated thin samples, the XY and YX samples had the highest stiffness. Overall, the average E_A for the coated thin samples was 41,859 MPa which was 94% higher than the average E_A of 21,611 MPa for the coated standard samples. Of the coated samples, the thin YZ orientation had the highest E_A (46,884 MPa) and the standard ZY orientation had the lowest E_A (19,443 MPa). When comparing the coated and uncoated samples, the coated thin sample average E_A value (41,859 Mpa) is over 30 times higher than the uncoated thin sample average E_A value (1258 MPa) and the coated standard sample average E_A value (21,611 MPa) is over 13 times higher than the standard sample uncoated average E_A value (1578 MPa). It should be noted that for coated samples, the thin samples had the highest E_A across all print orientations and for the uncoated samples, the opposite was true as the standard samples had the highest E_A across all print orientations. To support a clearer comparison between coated and uncoated sample specimen variability, the relative standard deviation was determined with the relative standard deviation being less than 10% for all thicknesses and all print orientations. ANOVA analysis indicated that there was a statistically significant difference of E_A based on orientation for all samples except the thin coated samples. Based on t tests statistical analysis, the standard coated samples were slightly less dependent on orientation with 7 pairs (XY–XZ, XY–YX, XY–YZ, XZ–YX, XZ–YZ, YX–YZ, and ZX–ZY) being statistically the same as opposed to the uncoated standard samples which had 5 pairs being statistically the same (XY–XZ XY–YX, XY–YZ, XZ–YX, and YZ–ZX). Uncoated thin samples also demonstrated high dependence on orientation with only 3 pairs (XZ–YZ, XZ–ZX, and YZ–ZX) showing no statistical change in E_A based on orientation. Overall, comparing print orientation and sample thickness, it is clear that sample thickness and coating had a larger impact on E_A than print orientation.

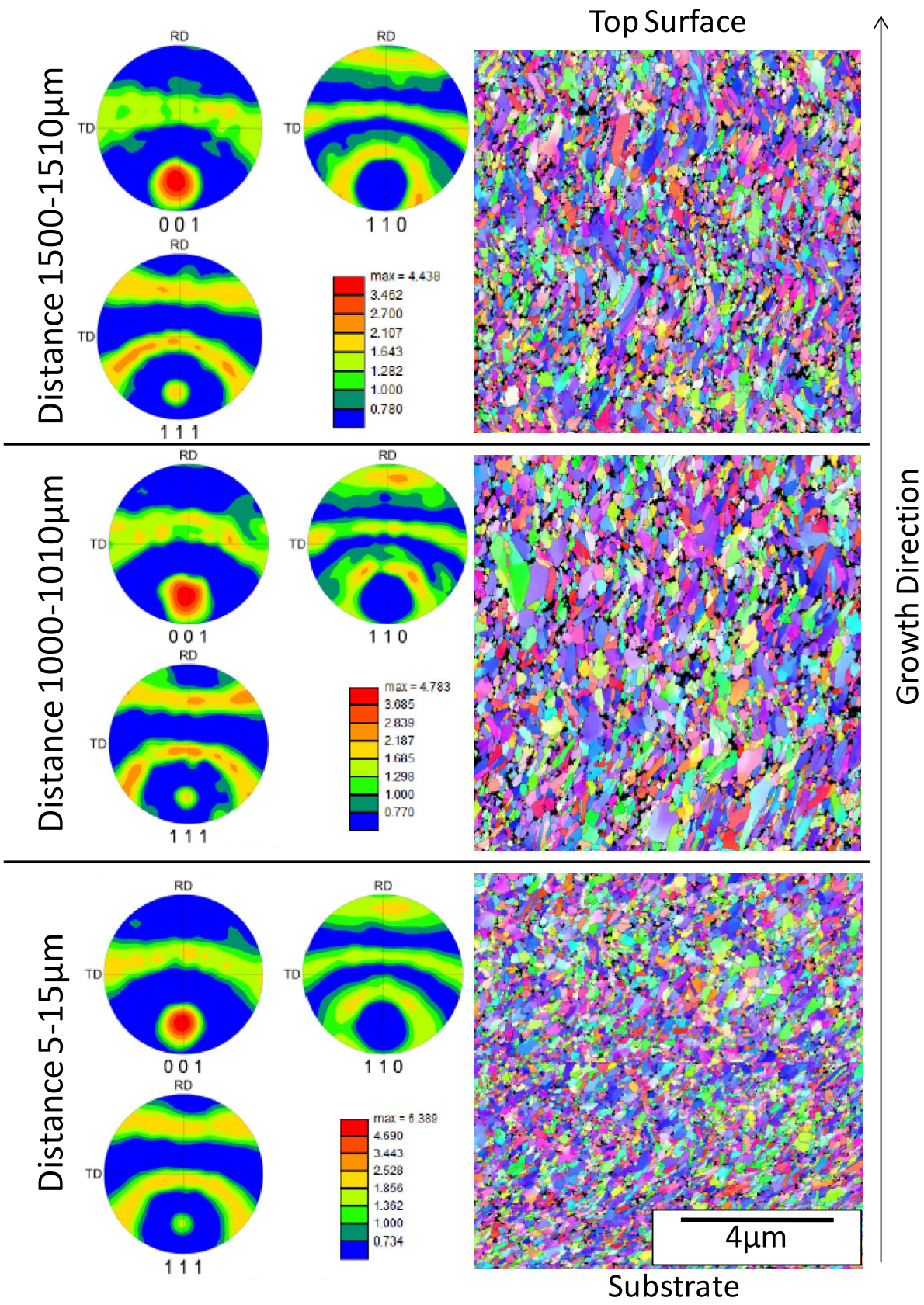


Fig. 6 EBSD plot of the NDPA nickel

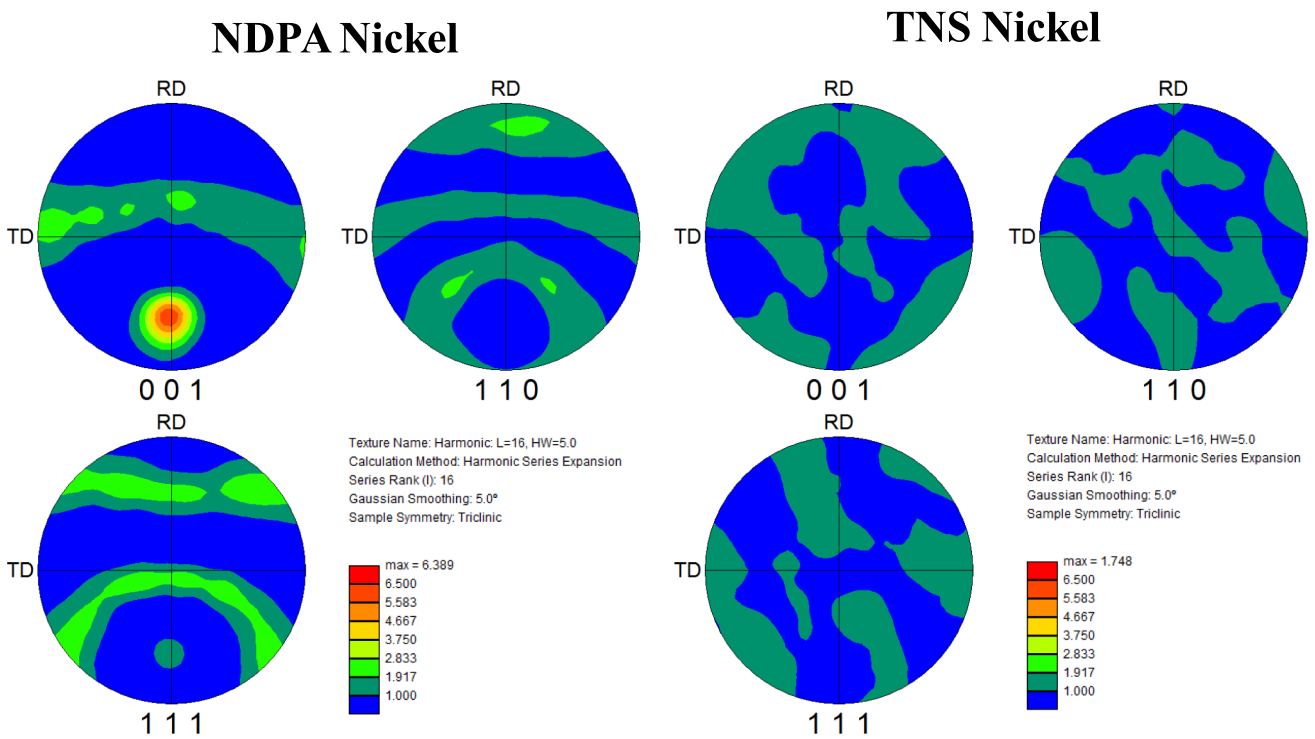


Fig. 7 Inverse pole figures of NDPA nickel and TNS nickel with the same linear scale for direct comparison. The NDPA nickel shows a strong < 201 > texture. The TNS sample shows no preferential texture

Table 3 Average axial elastic modulus (E_A) with standard deviation values for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

Orientation			XY	XZ	YX	YZ	ZX	ZY	Average
Coated	Standard	E_A (MPa)	22,316	22,638	22,822	22,408	20,041	19,443	**21,611**
		Std Dev	698 / 3.1%	287 / 1.3%	316 / 1.4%	928 / 4.1%	708 / 3.5%	301 / 1.6%	
	Thin	E_A (MPa)	41,415	43,115	41,690	46,884	42,150	35,899	**41,859**
		Std Dev	1099 / 2.7%	4119 / 9.6%	3117 / 7.5%	933 / 2.0%	3658 / 8.7%	2966 / 8.3%	
Uncoated	Standard	E_A (MPa)	1629	1646	1708	1540	1533	1411	**1578**
		Std Dev	4.0 / 0.2%	15.9 / 1.0%	38.0 / 2.2%	45.9 / 3.0%	24.2 / 1.6%	27.9 / 2.0%	
	Thin	E_A (MPa)	1384	1135	1547	1101	1131	1253	**1258**
		Std Dev	26.1 / 1.9%	48.9 / 4.3%	8.0 / 0.5%	34.7 / 3.2%	36.5 / 3.2%	19.9 / 1.6%	

3.3.2 Yield strength (σ_y)

A graphical representation of the effects of thickness, print orientation, and electroplate coating on σ_y is provided in Fig. 10 and Table 4 numerically presents the same results. This graph shows that the coated samples all had a significantly higher σ_y than their uncoated counterparts. For the coated standard samples, the XY, YX, and YZ print orientations had a slightly higher σ_y than the ZX and ZY print orientations, with only the XZ orientation falling slightly below the ZX orientation. For the coated thin samples, the XY, XZ, YX, and YZ samples all had a higher σ_y than the ZX and ZY print orientations. Overall, the average σ_y for the coated thin samples was 171.5 MPa which was 109% higher than the average σ_y of 82.1 MPa for the coated standard samples. Similar to the E_A results, the coated thin YZ sample had the highest σ_y (176.7 MPa) and the coated standard ZY sample had the lowest σ_y (68.8 MPa). When comparing the coated and uncoated samples, the thin sample coated average

Table 4 Average 0.2% yield strength (σ_y) with standard deviation values for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

Orientation			XY	XZ	YX	YZ	ZX	ZY	Average
Coated	Standard	σ_y (MPa)	88.9	79.8	88.7	84.4	82.3	68.8	**82.1**
		SD	3.5 / 3.9%	3.1 / 3.9%	1.1 / 1.2%	1.7 / 2.1%	2.9 / 3.5%	2.6 / 3.7%	
	Thin	σ_y (MPa)	172.1	174.7	174.7	176.7	164.6	166.2	**171.5**
		SD	9.9 / 5.7%	3.3 / 1.9%	1.1 / 0.6%	11.6 / 6.6%	8.6 / 5.2%	11.4 / 6.9%	
Uncoated	Standard	σ_y (MPa)	27.4	26.5	28.2	26.3	23.7	21.4	**25.6**
		SD	0.6 / 2.1%	0.5 / 1.9%	1.8 / 6.4%	0.4 / 1.4%	1.1 / 4.7%	0.4 / 1.8%	
	Thin	σ_y (MPa)	26.2	18.2	23.6	19.4	17.8	16.4	**20.3**
		SD	1.7 / 6.4%	1.2 / 6.6%	1.4 / 5.9%	1.5 / 7.9%	1.4 / 7.7%	1.0 / 5.8%	

Table 5 Average ultimate strength (UTS) with standard deviation values for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

Orientation			XY	XZ	YX	YZ	ZX	ZY	Average
Coated	Standard	UTS (MPa)	123.2	122.3	134.5	122.0	116.8	102.3	**120.2**
		SD	4.0 / 3.3%	3.4 / 2.8%	4.3 / 3.2%	1.8 / 1.5%	2.1 / 1.8%	3.8 / 3.7%	
	Thin	UTS (MPa)	251.9	240.2	251.7	246.8	226.2	221.3	**239.7**
		SD	4.6 / 1.8%	10.4 / 4.3%	5.2 / 2.1%	9.7 / 3.9%	6.0 / 2.7%	16.5 / 7.4%	
Uncoated	Standard	UTS (MPa)	44.7	44.6	43.6	43.1	32.3	28.7	**39.5**
		SD	0.7 / 1.6%	0.5 / 1.2%	1.1 / 2.6%	0.1 / 0.3%	2.1 / 6.5%	1.8 / 6.4%	
	Thin	UTS (MPa)	38.1	27.8	36.0	27.8	22.1	20.5	**28.7**
		SD	1.5 / 3.9%	1.4 / 5.1%	0.6 / 1.7%	1.5 / 5.5%	1.5 / 6.9%	0.6 / 3.1%	

Table 6 Average strain at failure (ϵ_f) with standard deviation values for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

Orientation			XY	XZ	YX	YZ	ZX	ZY	Average
Coated	Standard	$\boldsymbol{\varepsilon_f}$ (%)	1.56	1.68	2.09	1.60	1.65	1.62	**1.70**
		SD	0.13 / 8.4%	0.20 / 12.0%	0.15 / 7.3%	0.11 / 7.1%	0.09 / 5.3%	0.23 / 14.3%	
	Thin	ε_f (%)	1.78	1.46	1.68	1.61	1.43	1.39	**1.56**
		SD	0.30 / 16.6%	0.25 / 17.2%	0.05 / 3.1%	0.21 / 13.0%	0.12 / 8.7%	0.22 / 16.1%	
Uncoated	Standard	ε_f (%)	9.5	11.2	8.4	9.4	3.7	3.4	**7.62**
		SD	1.1 / 11.8%	1.0 / 9.1%	0.9 / 11.1%	1.2 / 12.4%	0.9 / 24.4%	0.9 / 25.1%	
	Thin	ε_f (%)	7.8	6.5	6.6	5.5	3.3	2.7	**5.40**
		SD	1.0 / 13.1%	0.6 / 9.2%	1.1 / 17.3%	0.8 / 15.2%	0.6 / 17.7%	0.1 / 2.2%	

σ_y value (171.5 MPa) is over eight times higher than the thin sample uncoated average σ_y value (20.3 MPa) and the standard sample coated average σ_y value (82.1 MPa) is a little over three times the standard sample uncoated average σ_y (25.6 MPa). In alignment with the results for E_A, the coated thin samples had the highest σ_y across all print orientations. The opposite was true for the uncoated samples which had the highest σ_y for the standard thickness samples across all print orientations. To support a clearer comparison between coated and uncoated sample specimen variability, the relative standard deviation was determined with the relative standard deviation being less than 8% for all thicknesses and all print orientations. ANOVA analysis again indicated that there was a statistically significant difference of σ_y based on orientation for all samples except the thin coated samples.

Table 7 Print Orientation Degree of Anisotropy (Note: uncoated data was previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

		Elastic Modulus (E_A)				Yield Strength (σ_y)			
		Uncoated		Coated		Uncoated		Coated	
Standard		Orientation	DoA	Orientation	DoA	Orientation	DoA	Orientation	DoA
	DoA (min)	YZ–ZX	0.4%	XY–YZ	0.4%	XZ–YZ	0.7%	XY–YZ	0.3%
	DoA (max)	YX–ZY	19.0%	YX–YZ	16.0%	YX–ZY	27.3%	XY–ZY	25.6%
	Avg		8.1%		7.8%		12.1%		10.8%
Thin		Orientation	DoA	Orientation	DoA	Orientation	DoA	Orientation	DoA
	DoA (min)	YX–YZ	0.4%	XY–YX	0.7%	XZ–ZX	2.2%	XZ–YX	0.0%
	DoA (max)	YX–YZ	33.7%	YZ–ZY	26.5%	XY–ZY	46.2%	YX–ZX	7.1%
	Avg		15.9%%		9.8%		21.5%		3.5%
		Ultimate Tensile Strength (UTS)	Strain at Failure (ε_f)						
		Uncoated	Coated	Uncoated	Coated				
Standard		Orientation	DoA	Orientation	DoA	Orientation	DoA	Orientation	DoA
	DoA (min)	XY–XZ	0.3%	XZ–YZ	0.2%	XY–YZ	0.6%	YZ–ZY	1.0%
	DoA (max)	XY–ZY	43.7%	YX–ZY	27.2%	XZ–ZY	106.4%	XY–YX	29.1%
	Avg		21.1%		10.2%		55.0%		10.8%
Thin		Orientation	DoA	Orientation	DoA	Orientation	DoA	Orientation	DoA
	DoA (min)	XY–YZ	0.1%	XY–YX	0.1%	YX-XZ	0.5%	XZ–ZX	1.8%
	DoA (max)	XY–ZY	60.1%	YX–ZY	12.9%	XY–ZY	97.8%	XY–ZY	24.4%
	Avg		30.0%		6.6%		48.1%		12.0%

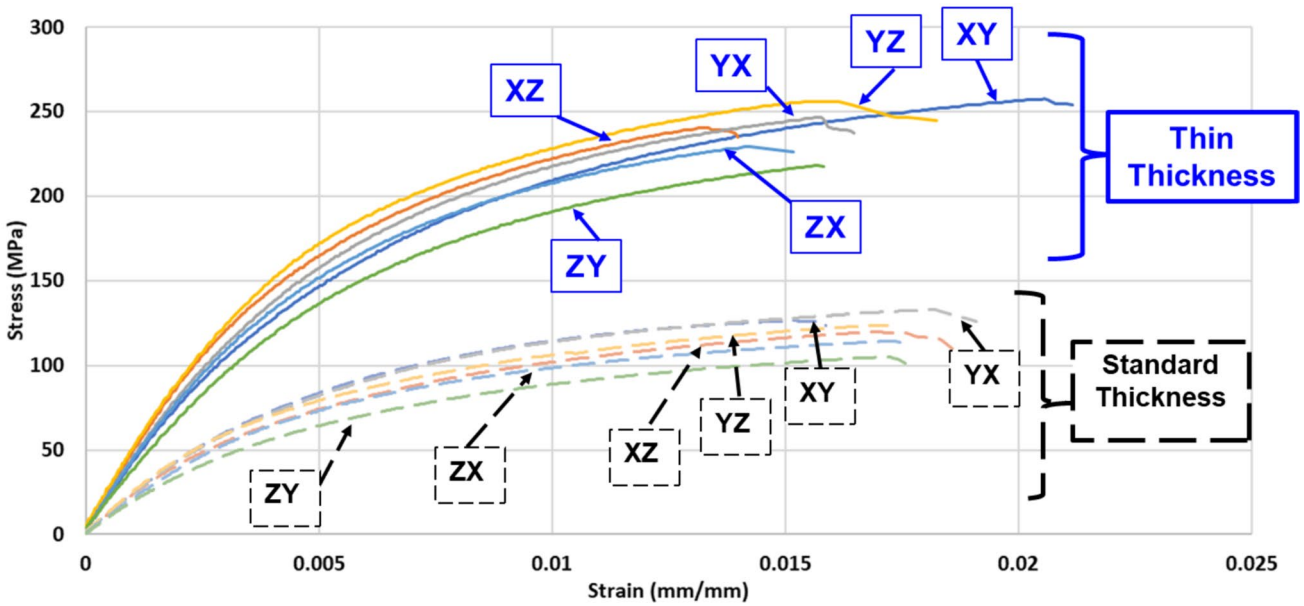


Fig. 8 Representative stress–strain curves of thick and thin coated tensile specimens at orthotropic print orientations

Based on t tests statistical analysis, the standard uncoated, standard coated, and thin uncoated samples all had 11 orientation pairs being statistically different. However, the pairs were different for each sample type. Standard uncoated orientation pairs XY–XZ, XY–YX, XY–YZ, XZ–YX, XZ–YZ, and YX–YZ had no statistical difference in σ_y based on orientation. Standard coated orientation pairs XY–YX, XY–YZ, XY–ZX, XZ–YZ, XZ–ZX,

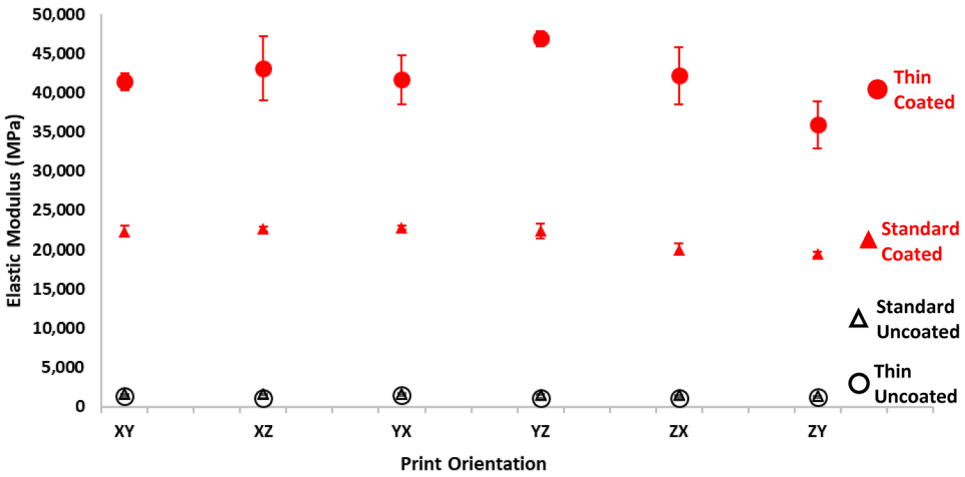


Fig. 9 Average axial elastic modulus (E_A) with standard deviation bars for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

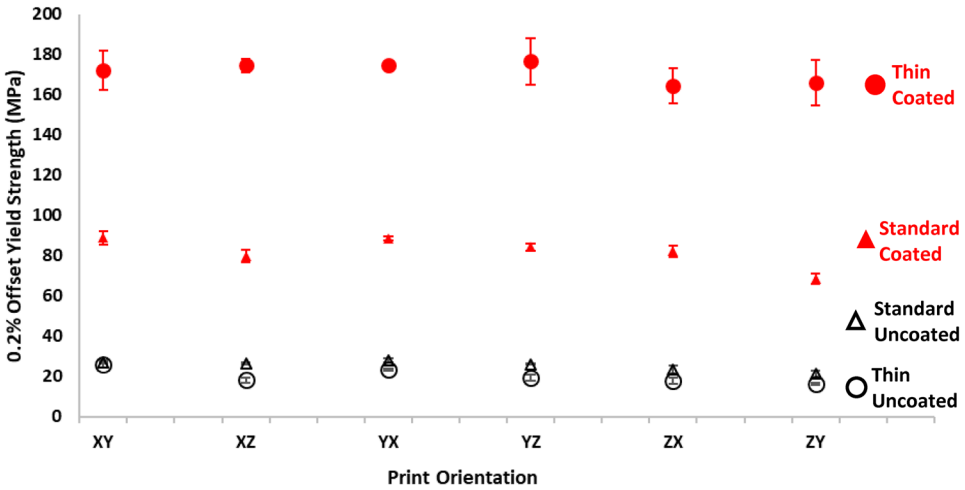


Fig. 10 Average 0.2% yield strength (σ_y) with standard deviation bars for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

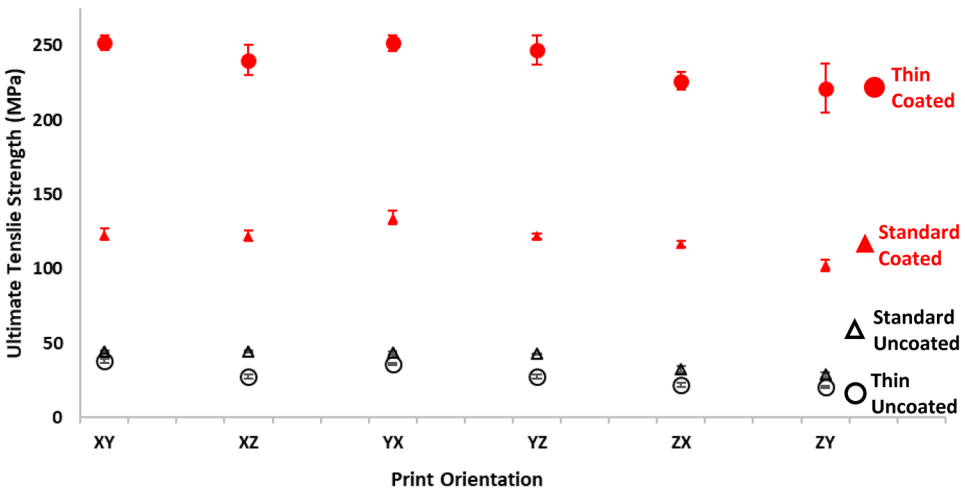


Fig. 11 Average ultimate tensile strength (UTS) with standard deviation bars for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

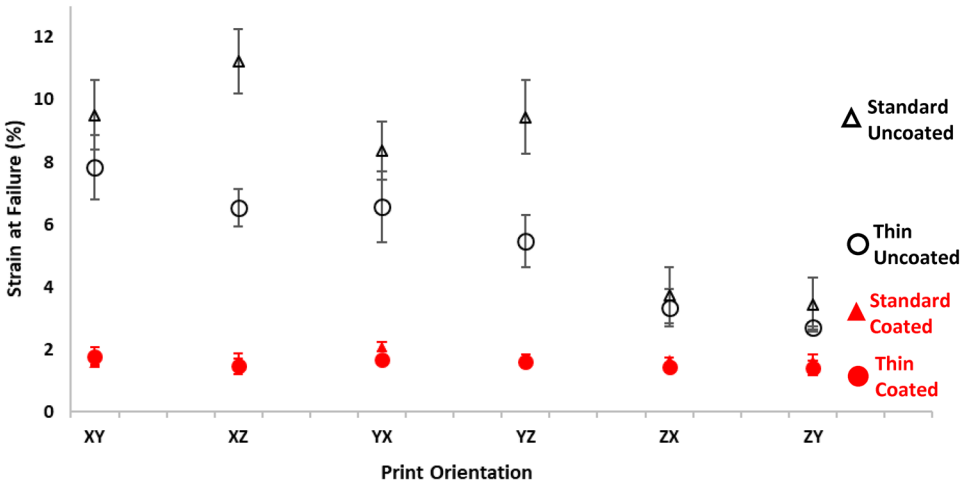


Fig. 12 Average strain at failure (ϵ_f) with standard deviation bars for coated and uncoated specimens of both standard and thin thickness at all orthotropic print orientations. (Note: uncoated data were previously published by Slager et al. [10] and shown here to provide a direct comparison between the coated and uncoated samples)

and YZ–ZX had no statistical difference in σ_y based on orientation. Thin uncoated orientation pairs XY–YX, XZ–YZ, XZ–ZX, XZ–ZY, YZ–ZX, and ZX–YY had no statistical difference in σ_y based on orientation. Similar to E_A results, when comparing print orientation and sample thickness, it is clear that sample thickness and coating had a larger impact on σ_y than print orientation.

3.3.3 Ultimate tensile strength (UTS)

A graphical representation of the effects of thickness, print orientation, and electroplate coating on UTS is provided in Fig. 11 and Table 5 numerically presents the same results. This graph shows that the coated samples all had a significantly higher UTS than their uncoated counterparts. For both the standard and coated thin samples, the XY, XZ, YX, and YZ print orientations all had a higher UTS than the ZX and ZY print orientations. Overall, the average UTS for the coated thin samples was 239.7 MPa which was 99% higher than the average UTS of 120.2 MPa for the coated standard samples. For the coated samples, the thin XY orientation had the highest UTS (251.9 MPa) with the all of the X and Y print orientations having similar UTS values and the standard ZY orientation had the lowest UTS (102.3 MPa). When comparing the coated and uncoated samples, the coated thin sample average UTS value (239.7 MPa) was over eight times higher than the uncoated thin sample average UTS value (28.7 MPa) and the coated standard sample average UTS value (120.2 MPa) was a little over three times the uncoated standard sample average UTS (39.5 MPa). In alignment with the results for E_A and σ_y, the coated thin samples had the highest UTS across all print orientations and for the uncoated samples. Conversely, uncoated standard thickness samples exhibited higher UTS values across all print orientations compared to uncoated thin samples. To quantify specimen variability between coated and uncoated samples, the relative standard deviation was calculated and found to be less than 8% across all thicknesses and print orientations. ANOVA analysis again indicated that there was a statistically significant difference of UTS based on orientation for all samples except the thin coated samples. Based on t tests statistical analysis, the standard samples were less dependent on orientation with 5 pairs for both uncoated (XY–XZ, XY–YX, XZ–YX, YX–YZ, and ZX–ZY) and coated (XY–XZ, XY–YZ, XY–ZX, XZ–YZ, XZ–ZX) having no statistical difference as opposed to the uncoated thin samples which only had 3 pairs with no statistical difference (XY–YX, XZ–YZ, and ZX–ZY). Similar to E_A and σ_y results, when comparing print orientation and sample thickness it was clear that sample thickness and coating had a larger impact on UTS than print orientation.

3.3.4 Strain at failure (ϵ_f)

A graphical representation of the effects of thickness, print orientation, and electroplate coating on ϵ_f is provided in Fig. 12 and Table 6 numerically presents the same results. This graph shows that the coated samples all had a significantly lower ϵ_f than their uncoated counterparts. In general, ϵ_f is not significantly different for the various print orientations of the coated samples. Overall, the average ϵ_f for the coated thin samples was 1.57%, which was only slightly less than the average ϵ_f of 1.70% for the coated standard samples. When comparing the coated and uncoated samples, the coated thin sample average ϵ_f value (1.57%) was over three times lower than the uncoated thin sample average ϵ_f value (5.40%) and the coated standard sample average ϵ_f value (1.70%) was over four times lower than the uncoated standard sample average ϵ_f (7.62%). To support a clearer comparison between coated and uncoated sample specimen variability, the relative standard deviation was determined with the relative standard deviation being less than 30% for all thicknesses and all print orientations. This was a substantially higher relative standard deviation variance than

Table 8 Comparison of experimentally measured and ROM predicted tensile properties

Orientation			XY	XZ	YX	YZ	ZX	ZY	Average
E_A (MPa)	Standard	Predicted	22,817	22,830	22,881	22,743	22,737	22,636	**22,774**
		Measured	22,316	22,638	22,822	22,408	20,041	19,443	**21,611**
		% Difference	2.2%	0.9%	0.3%	1.5%	12.6%	15.2%	5.2%
	Thin	Predicted	45,193	45,034	45,298	45,012	45,031	45,109	**45,113**
		Measured	41,415	43,115	41,690	46,884	42,150	35,899	**41,859**
		% Difference	8.7%	4.4%	8.3%	-4.1%	6.6%	22.7%	7.48%
σ_y (MPa)	Standard	Predicted	146.6	145.9	147.2	145.7	143.5	141.6	**145.1**
		Measured	88.9	79.7	88.7	84.3	82.3	68.8	**82.1**
		% Difference	48.9%	58.6%	49.6%	53.4%	54.2%	69.3%	55.4%
	Thin	Predicted	272.6	267.5	271.0	268.3	267.2	266.3	**268.8**
		Measured	172.1	174.7	174.7	176.7	164.6	166.2	**171.5**
		% Difference	45.2%	42.0%	43.2%	41.1%	47.6%	46.3%	44.2%
UTS (MPa)	Standard	Predicted	195.4	195.3	194.5	194.1	185.1	182.2	**191.1**
		Measured	123.2	122.3	134.4	122.0	116.8	102.3	**120.2**
		% Difference	45.3%	46.0%	36.5%	45.6%	45.3%	56.2%	45.6%
	Thin	Predicted	351.5	344.8	350.1	344.8	341.2	340.1	**345.4**
		Measured	251.9	240.2	251.7	246.8	226.1	221.3	**239.7**
		% Difference	33.0%	35.8%	32.7%	33.1%	40.5%	42.3%	36.1%

the other material properties presented. ANOVA analysis again indicated that there was a statistically significant difference of ϵ_f based on orientation for all samples except the thin coated samples. Based on t tests statistical analysis uncoated samples, both had 9 orientation pairs being statistically different. However, not all pairs were the same for each sample type. Standard uncoated orientation pairs XY–XZ, XY–YX, XY–YZ, XZ–YZ, YX–YZ, and ZX–ZY had no statistical difference in ϵ_f based on orientation. Thin uncoated orientation pairs XY–XZ, XY–YX, XZ–YX, XZ–YZ, YX–YZ, and ZX–ZY had no statistical difference in ϵ_f based on orientation. Standard coated orientation pairs were much less dependent on orientation with only 5 pairs indicating a difference in ϵ_f based on orientation (XY–YX, XZ–YX, YX–YZ, YX–ZX, and YX–ZY). While sample thickness and coating generally had a larger impact on ϵ_f, it was observed that the ZX and ZY print orientations had the lowest ϵ_f for most samples.

3.4 Coated sintered PA12 specimen print orientation degree of anisotropy results (D_oA)

D_oA was calculated using Eq. (3) and can be found in Table 7. In all cases, the D_oA decreased for the coated specimens compared to the uncoated. The greatest average decrease in D_oA from uncoated to coated based on property was found for ϵ_f. There was less of a decrease in D_oA with less plastic deformation: smallest change for E_A and largest change in ϵ_f from coated to uncoated. Based on thickness, the thin specimens had a greater decrease in D_oA than the standard parts from uncoated to coated.

3.5 Coated sintered PA12 material ROM property prediction results

ROM (Eqs. 1, 2) was used to calculated predicted values of E_A, σ_y, and UTS and compared to the tested values discussed in Section 3.2. Measuring the gauge cross-section before and after plating resulted in a 17.3 and 35.7% coated material for the standard and thin specimens, respectively. For all coatings used, the metallic coating was vastly stiffer and stronger than the PA12, and so, the ROM results were largely driven by the coating properties. The predicted values (calculated using the ROM equation), with a comparison to the experimental values, are listed in Table 8. E_A predictions match much better than σ_y and UTS. The E_A prediction was within 10% for orientations on the build plate and went up to 23% for the thin ZY orientation. However, overall E_A prediction average was well under 10% for both the standard and thin specimens. Thin specimens predicted material values matched the experimental values closer than the thick specimens. σ_y and UTS predictions were much less accurate than E_A predictions with an average deviation of 36–54%. UTS predictions were slightly better than σ_y and thin specimen predictions were slightly better than standard specimen predictions. E_A data (property well within the elastic regime) indicate the feasibility of using ROM to predict material properties. However, σ_y and UTS data do not correlate well enough with ROM calculations to effectively predict these properties using ROM.

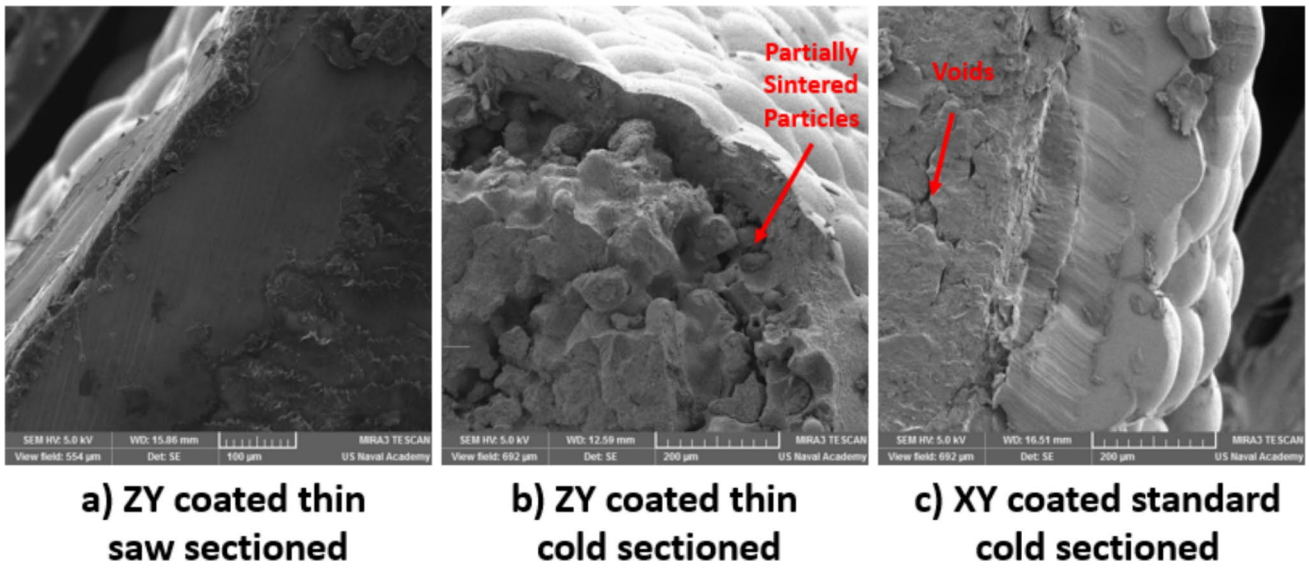


Fig. 13 SEM 100×and 200×of sectioned coated samples using **a** slow speed diamond blade saw at room temperature and **b**-**c** razor blade wedge after soaking specimen in liquid nitrogen

3.6 Coated sintered PA12 cross-sectioned microscopy results

Specimens were sectioned to compare pre-fracture cross-section (Fig. 13) and fractured cross-section (Figs. 14 and 15). The process of sectioning tended to damage the cross-section and influence the results, so multiple methods were used to support characterization. Using a slow speed circular diamond blade at room temperature (Fig. 13a) and a razor blade pushed through the section as a wedge after soaking the sample in liquid nitrogen (Fig. 13b-c) resulted in the least amount of cross-section and material interface damage during sectioning. The sectioning done at room temperature did not indicate any separation at the PA12/coating interface; however, plastic deformation ("smearing") of the PA12 made it difficult to identify the interface. Sectioning at cryogenic temperatures (Fig. 13b-c) mitigated most of the plastic deformation during sectioning but did appear to cause some damage that may have pulled the PA12 away from the coating. Figure 13c also appears to show three distinct layers (electroplating materials) in the coating. In all cases, it does appear the coating filled in most of the rough particulate SLS surface and bonded to the PA12 core. Though voids and under-sintered particles are evident in both of the cryogenic sections (Fig. 13b-c), these print defects are much more pronounced in the thin section (Fig. 13b). Globular hemisphere-like growth on the surface of the coating can be seen on all sections. The size of the globular shapes is much larger than the PA12 particle size and is consistent with other work describing these globular features during electroplating [17]. In both images, the coating appears to fill in the particulate surface of the printed part as well as fill in any edge defects.

3.7 Coated sintered PA12 specimen fractography results

The tensile specimen fracture surfaces were analyzed using SEM to identify and evaluate factors that may have contributed to the evaluated material properties and anisotropy. The internal sintered PA12 specimens were imaged to qualitatively identify voids, porosity, and particle adhesion/sintering. The uniformity, adhesion, and geometry of the coating were also analyzed to investigate the interaction between the sintered PA12 and electroplated metal coating during tensile loading. Figure 14 is a fracture image at 100x magnification of the coated standard samples at each different print orientation and Fig. 15 is a fracture image at 100x magnification of the coated thin samples at each different print orientation.

All fracture surfaces were relatively flat and appear to have little plastic deformation at the fracture surface. Standard samples orientation ZX and XZ (Fig. 14) have some areas of what appears to be ductile cup and cone fracture, but still with very little overall plastic deformation. Thin samples indicate similar areas that are most pronounced in YX, ZX, and YZ (Fig. 15), but the formations appear to be more stretched fibrils than cup and cone. Though the overall fracture mode appears to be relatively brittle for both the sintered PA12 and the metal coating, when compared to the sectioned samples (Fig. 13), it does appear the sintered PA12 core elastically deformed more than the metal coating which caused debonding at the PA12/metal interface. The

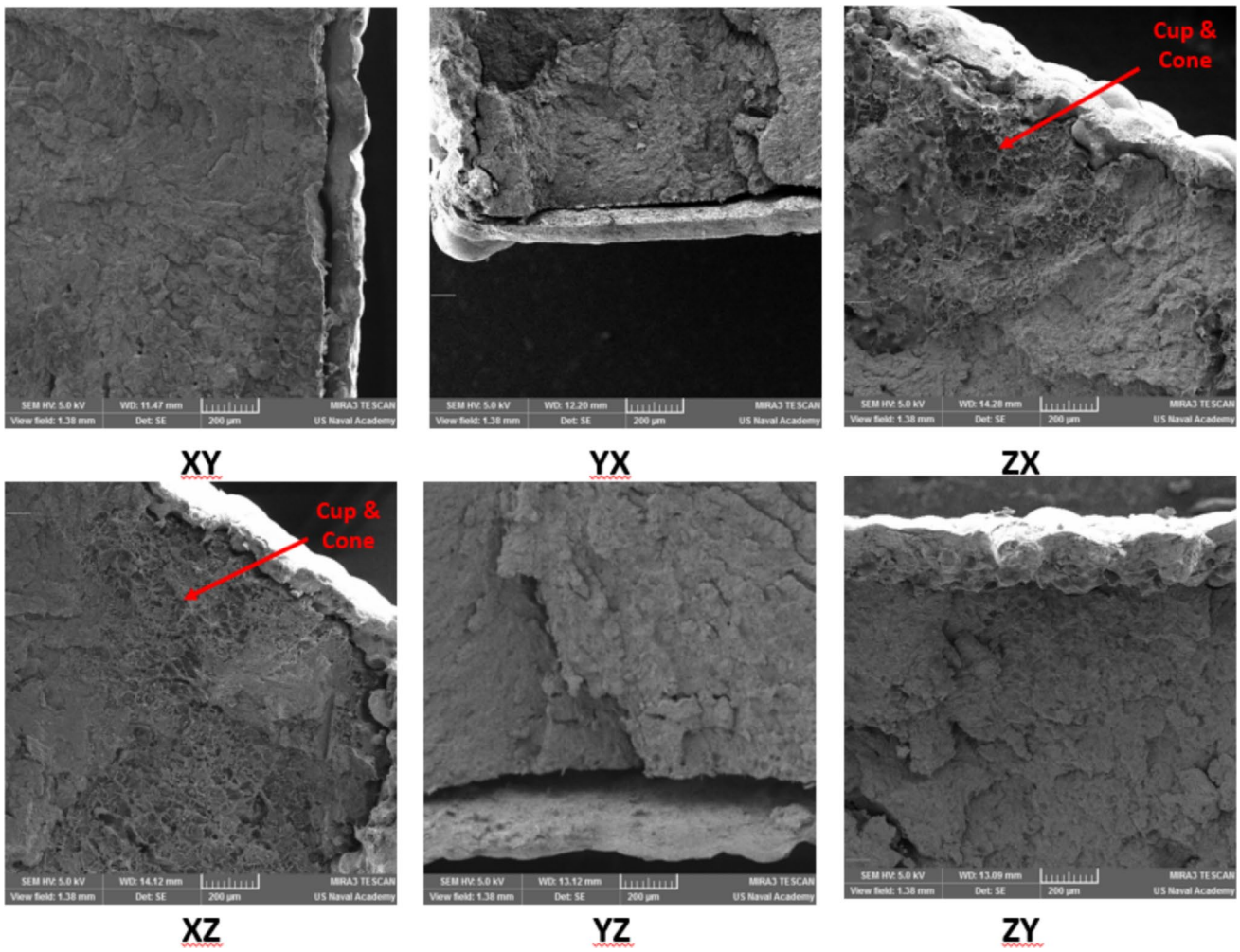


Fig. 14 SEM 100× magnification of coated standard samples

debonding and pulling away of the PA12 from the metal coating was less pronounced in the Z orientation (ZX and ZY) for both standard and thin samples. In general, the overall interface debonding also appeared to be less pronounced in the thin samples than the standard samples. Some of the thick samples (XY and ZX) also appear to have some debonding within the metal coating.

Consistent with the sectioned images (Fig. 13), the standard samples (Fig. 14) appear to have more uniform sintering with no noticeable unsintered particles. The thin samples (Fig. 15) in comparison tend to have more sintering defects (noticeably porosity and under-sintered particles). Globular formations on the surface of the coatings are also present (similar to the sectioned specimens).

4 Discussion and future direction

This research demonstrates a novel hybrid materials approach combining SLS with strategic electroplating to dramatically enhance part performance while mitigating traditional SLS limitations. Through a novel electroplating process using copper and two distinct nickel depositions, this work accomplished improvements in mechanical properties—including elastic modulus and yield strength, while reducing print orientation anisotropy.

Mechanical characterization of electroplating microsamples with a footprint of 3 mm x 1 mm and a gauge section of 250 µm x 250 µm allow investigation into the properties in the growth direction as well as the transverse direction. It was found that the copper deposited FCC material had no directional dependence and behaved very isotropically, while the nickel deposited materials had variations in the two directions tested. In addition, both materials showed significant strengthening when compared to other manufacturing

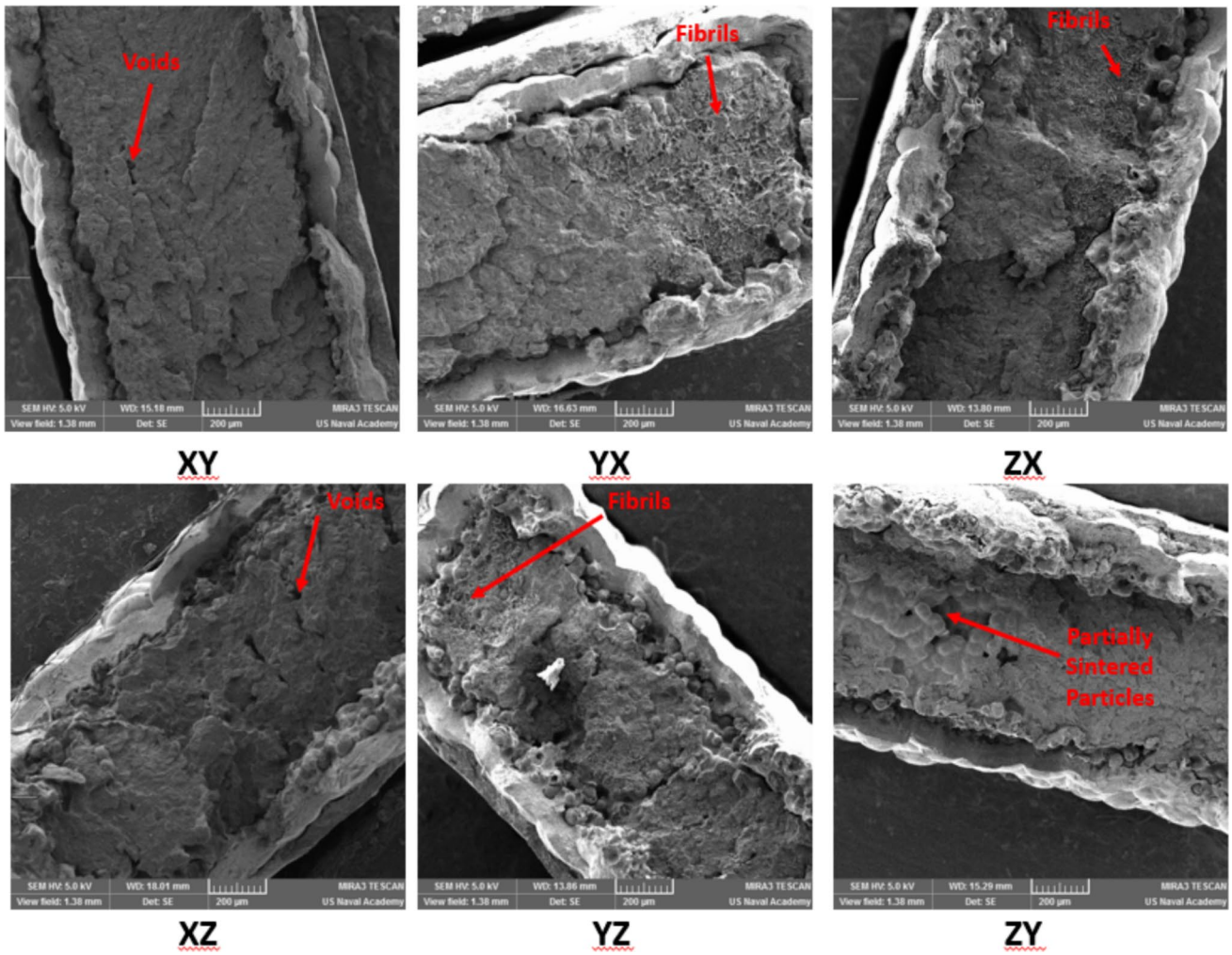


Fig. 15 SEM 100× magnification of coated thin samples

processes, such as casting or extrusion of the base metals. This was linked to grain refinement down to sub-100 nm for the deposition process. Nickel materials that were deposited with supplemental platinum anodes have strength exceeding two times that of traditionally electrodeposited nickel. A significant number of grains in the NDPA nickel measure below 50 nm with the possibility to reduce size further through precise control of the supplemental anodes. The justification for the strengthening was related back to the grain structure of the materials with anisotropy being related to the texture of the deposits. The reason for the varying properties of the coatings for the nickel materials was related back to slight residual compressive stress in the nickel deposits as well as grain texture and evolution. The grains grow larger in diameter as they increase in distance from the substrate. This results in a higher propensity for larger weaker grains as material deposits become thicker. Grain texture in the growth direction heavily favors the $< 201 >$ directions, which also explains the change in properties of the growth direction as only 2 of the 12 FCC slips systems are active for the NDPA nickel. The properties measured are very useful for functionalization of electrodeposited structures formed from 3D-printing, as the base substrate can be designed to take advantage of the unique texturing that occurs within the material. While ROM can serve to guide the structural enhancement of coated polymers, finite-element optimizations can be varied for directional properties as well as thickness-dependent properties. It is hypothesized that diminishing returns will occur for electrodeposited materials, since the grain size is linked to the thickness of the coating. Iterative processes will most likely be required to optimize mechanical structures using electrodeposition as the primary strengthening phase.

Reductions in strength for the growth direction NDPA nickel can be attributed to grain evolution. Grains approximately 250 μm from the substrate have an average size of 74 nm, while grains 1.5 mm from the substrate are 120 nm in diameter. The samples extracted from this portion fail in

the gage section; however, they fail near the thicker section of the deposit. This is due to larger gains present in this area. The resultant strength of the material is much lower at the larger grain size. The cause of the increase in grain size is attributed to columnar grains forming within the deposit. These grains tend to grow in diameter as they progress. The columnar grains do not span the full length of the deposit but grow in a columnar fashion for a brief period. The phenomenon regulating this is not yet well understood but is postulated to be related to the grain surface energy of non-columnar grains.

Increasing both tensile properties and isotropy hinged on achieving uniform and strong adhesion between the non-metallic printed material and the metallic coating. The high throw acid copper bath appeared to place relatively complex geometries (Fig. 1) while filling the voids of the SLS powder surface (Fig. 13). The initial acid copper bath formulations used aided in coating corners and slots and provided good metal filling into the relatively rough SLS surfaces. This was additionally improved by the slow plating rates used which appears to also aid in bridging copper across fine gaps. These findings show promise of adding electroplating to SLS design for AM without incorporating expensive and extensive post-processing such as vapor smoothing. Further investigation is still warranted to find techniques that increase coating uniformity and adhesion while decreasing globular formations on exterior surfaces.

Tensile results of the coated sintered PA12 specimens showed a significant increase in tensile properties from uncoated to coated as well as a decrease in anisotropy. Similar to previous research done by Slager et al. [10], inconsistencies in sintering are more pronounced in the thin samples; however, electroplating filled in the edge defects and reduced the D_oA due to print orientation. Sample thickness and coating had a larger impact than print orientation and average DoA was reduced for all tensile properties when the specimen was coated. E_A for standard specimens experienced the smallest increase in isotropy with coating. However, coating significantly decreased DoA due to orientation for all other specimens and material properties. Most properties benefited from at least a 50% decrease in DoA and strain at failure saw the largest decrease in DoA going from approximately 50% uncoated to around 10% coated (Table 7). While the ANOVA and t test analysis did not definitively identify the impact of orientation of print orientation on coated samples with the samples used, the incorporation of D_OA analysis more clearly highlighted these impacts. Future research with an increased sample size (larger than the three used for this study) could provide further insight into whether a difference more definitely exists.

Overall, fractography and sectioning indicated the separation between the PA12 and metal coating occurred during plastic deformation. Z orientations tend to be more brittle [10] resulting in less debonding during fracture. This orientation also experienced some of the greatest benefit in strength and mitigated some of the disadvantage of stressing a part in the build direction. Still, the results do indicate that the difference in stiffness between the metal and the PA12 will likely cause debonding if the sample is deformed well into the plastic regime (σ_y and UTS, respectively). It appears the plastic deformation and necking of the PA12 core pulled the PA12 away from the coating during plastic deformation; however, this alone does not appear to account for the 41–48% difference between measured and predicted σ_y. This is especially true considering the predicted values were slightly closer to measured values for UTS than σ_y. The difference in stiffness in the three different metal coatings could also cause some of the debonding within the metal coating; however, additional research is required to verify this, since it is not seen across all samples. It should be noted that sectioning electroplated polymer samples without causing the coating to de-bond from the polymer core while minimizing plastic deformation of the polymer core was exceptionally difficult. The process of sectioning tended to damage the cross-section (particularly for the PA12) and influence the sectioning results. Techniques to accomplish this were also not found in the literature and future work should build on the cryogenic sectioning work in this research to find ways to section samples without damaging and altering the sample in the process.

Other benefits of electroplating SLS parts, such as improved bending properties and environmental resilience, should be investigated in the future. Future work should build on this work and previous work done by Baligidad and others [8, 17, 29–31] to further investigate post-processing and surface preparation techniques on the printed SLS part to increase electroplate bonding. Increased PA12/metal bonding is expected to further decrease DoA while increasing final coated part mechanical properties. Krupova et al. [17] have found that this may also decrease the globular growths during the electroplating process.

Promising research avenues emerge from this study. This research also opens the door to customize SLS part properties through electroplating. The ROM can be used to determine the required electroplating material and thickness to achieve specific stiffness. However, the disparity between the ROM predictions and experimental results for σ_y and UTS (properties further in the plastic range) is particularly complex for sintered PA12 material. Building a reinforcing layer directly on a relatively weak core material has many benefits, but is complicated by the interaction of the two materials. One of the complications is the rough SLS surface. Many as-printed SLS surfaces have a transition from a well consolidated material at about 50–75 μm below the surface to a less-consolidated material at the surface [15]. The electroplating process, in this case, relies on the copper to

coat the low-density area and possibly offer some smoothing before the nickel builds a good reinforcing layer. A second complication between the core PA12 material and the metal coating is the difference in stiffness. The metal coating is often > 70 times stiffer than the plastic core. The difference in strength and stiffness is one of the primary advantages of coating SLS parts, but the present electroplating research has found that thin electroplated layers on soft substrates do not reinforce as well as thin electroplated layers on more rigid materials. The difference in stiffness and strength may cause some delamination at higher strains, but the mismatch in stiffness also often plastically loads the metal coating before the polymer core. Typically, at < 2% strain, the metal is fully loaded and the polymer core is only at ~ 20% of the load-carrying capacity. This can lead to the internal polymer elastically elongating and taking a small fraction of the load until the metal fractures, at which point the load is instantaneously transferred to the polymer resulting in immediate failure of the total structure. This phenomenon essentially results in sequential plastic regime loading of first the metal coating and then the polymer core. However, this research shows that this phenomenon may actually be swapped for some SLS build orientations resulting in the core PA12 failing before the metal coating. This highlights the importance of additional research when coating SLS parts because of the print orientation dependent material properties. Further investigation is needed to understand and optimize the difference between mechanical testing determined and predicted values for σ_y (41–48%) and UTS (33–42%), potentially through improved surface preparation, interface analysis, and techniques to improve and control metal–polymer bonding. Developing electroplating methods to ensure thickness uniformity around different part features especially between thin features (such as the gauge section in the Type-V tensile specimen) and thicker features (such as the grip section in the tensile specimen) may increase the predictability of total parts.

5 Conclusion

This research establishes a novel approach to addressing fundamental limitations in polymer additive manufacturing through strategic electroplating. By developing a novel three-layer metallic coating process, we have not only achieved unparalleled increases in mechanical properties but also effectively mitigated the long-standing challenge of print orientation anisotropy for thin samples (0.8 mm) across all tested material properties and orientations indicated by a low D_OA (3.5–12.0%) and ANOVA analysis showing no statistical difference in material properties based on orientation. Standard samples also showed a noticeable decrease in anisotropy for plastic deformation properties as indicated with average D_OA decrease from 21.1% to 10.2% for UTS and 55.0% to 10.8% for ϵ_f. This also highlights the importance of incorporating D_OA analysis and not simply using more classical methods such as ANOVA and t tests as the decrease in anisotropy was not immediately apparent when using these methods. Overall, though both AM and electroplating are inherently anisotropic, if employed together correctly, they can decrease the anisotropy of the final multi-material part.

The collected data also demonstrate the significance of electroplating thin and thick PA12 SLS specimens to improve tensile properties. Electroplated samples' tensile properties (E_A, σ_y, UTS) were all significantly higher than uncoated samples with the properties of the coating driving overall performance (E_A increased 1270% for standard and 3230% for thin samples, σ_y increased 220% for standard and 750% for thin samples, and UTS increased 200% for standard and 740% for thin samples). The increase in E_A, σ_y, and UTS did come at the cost of decreased ε_f which is attributed to the stiffer nature of the coating as compared to the base material (ε_f decreased 350% for standard and 250% for thin samples). When comparing print orientation and sample thickness, it is clear that sample thickness and coating had a larger impact on E_A, σ_y, and UTS than print orientation. Strain at failure had the opposite effect, but results varied less based on orientation when compared to uncoated samples. The success of this approach, particularly with thin-walled structures, demonstrates that the inherent trade-off between lightweight design and mechanical performance in polymer AM can be overcome through controlled metallization, while maintaining the cost and weight advantages that makes polymer AM attractive.

The implications of this work extend beyond the immediate mechanical property improvements measured and reported. Electroplating also opens the door to customizing part properties by calculating required coating thickness using ROM, which was demonstrated for E_A with an average deviation from ROM predicted values to measured values of 5.2% for standard samples and 7.5% for thin samples. ROM was not as accurate of a material property prediction tool for properties further in the plastic region, such as σ_y, UTS, and ε_f. suggesting that further research is prudent to expand this predictive capability beyond E_A. Overall, the study demonstrated that it is possible to create hybrid disparate phase polymer–metal components with predictable and tunable properties through control of electroplating parameters and grain structure refinement. Property tuning using electroplated polymer AM enables a new design paradigm to selectively enhance lightweight polymer structures to

achieve performance requirements, geometric complexity, and cost advantages.

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Declarations

Competing interest The authors declare no competing interests.

Data availability No datasets were generated or analyzed during the current study.

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